

VERRUCAE (VULGARIS AND PLANTAR WARTS)

February 2026 version





DISCLAIMER

- While progressing through this learning and using any of the resources provided, please keep in mind:
 - ✓ the scope of practice for pharmacists within your province or territory
 - ✓ the policies and procedures of your designated place of work
 - ✓ your scope of practice may be more limited or extensive than presented in this eLearning module
- The treatment examples provided are intended to illustrate possible indicated options for managing the specified common/minor ailment or condition. Pharmacists should:
 - ✓ follow provincial/territorial regulations regarding treatment options they are authorized to prescribe
 - ✓ exercise professional judgment at all times to prescribe the most appropriate option for each patient
 - ✓ refer to the latest clinical guidelines and product monographs

LEARNING OBJECTIVES

VERRUCAE (VULGARIS AND PLANTAR WARTS)



Describe the pathophysiology and clinical presentation of verrucae, including verruca vulgaris and verruca plantaris



Identify common risk factors for the development of verrucae



Identify differential diagnoses and red flags that may warrant referral



Understand the assessment of a patient presenting with suspected verrucae



Discuss non-pharmacological and pharmacological treatment options for verrucae



Identify side effects and considerations for specific treatments



Understand monitoring parameters and treatment timelines



LEGEND

DNCB: Dinitrochlorobenzene

DPCP: Diphenylcyclopropenone

HIV: Human Immunodeficiency Virus

HPV: Human Papillomavirus

PVD: Peripheral vascular disease

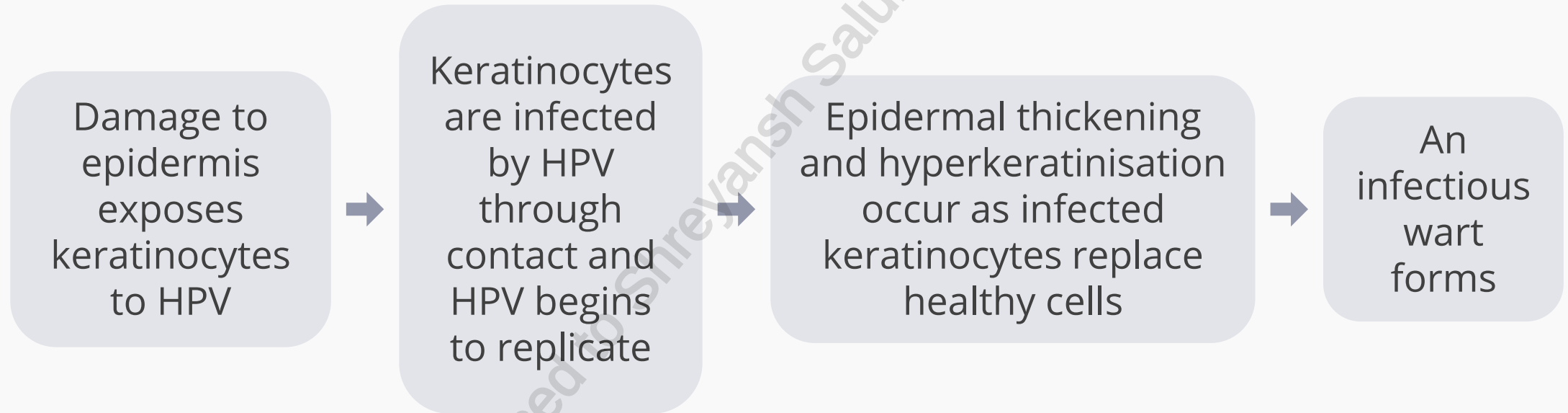
SADBE: Squaric Acid Dibutyl Ester

A top-down view of a doctor's desk. A person in a white lab coat is typing on a silver laptop. To the right, a blue stethoscope rests on a white clipboard with a blue folder underneath. A smartphone lies on the desk next to the laptop. In the foreground, a silver keyboard is visible. The desk is covered with various medical and office supplies, including a pen, glasses, and a small container. The background is a light-colored wooden surface. The text 'CLINICAL PRESENTATION & ASSESSMENT' is overlaid on the left side of the image.

CLINICAL PRESENTATION & ASSESSMENT



PATHOPHYSIOLOGY





CLINICAL PRESENTATION

The 3 most common verrucae, or cutaneous warts are:



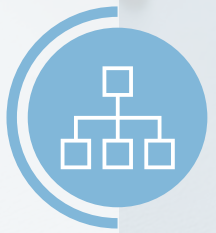
1. Common warts
(verruca vulgaris)



2. Plantar warts
(verruca plantaris)



3. Flat/plane warts
(verruca plana)



HPV STRAINS

HPV STRAIN	MANIFESTATION
HPV 1	Palmar and plantar warts (verruca plantaris)
HPV 2 and 4	Common warts (verruca vulgaris)
HPV 3 and 10	Flat or plane warts (verruca plana)



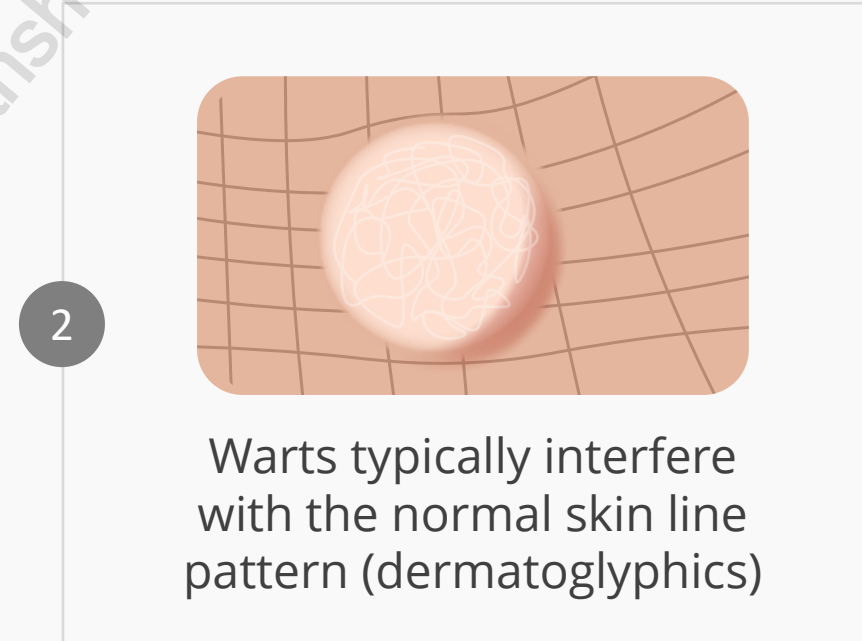
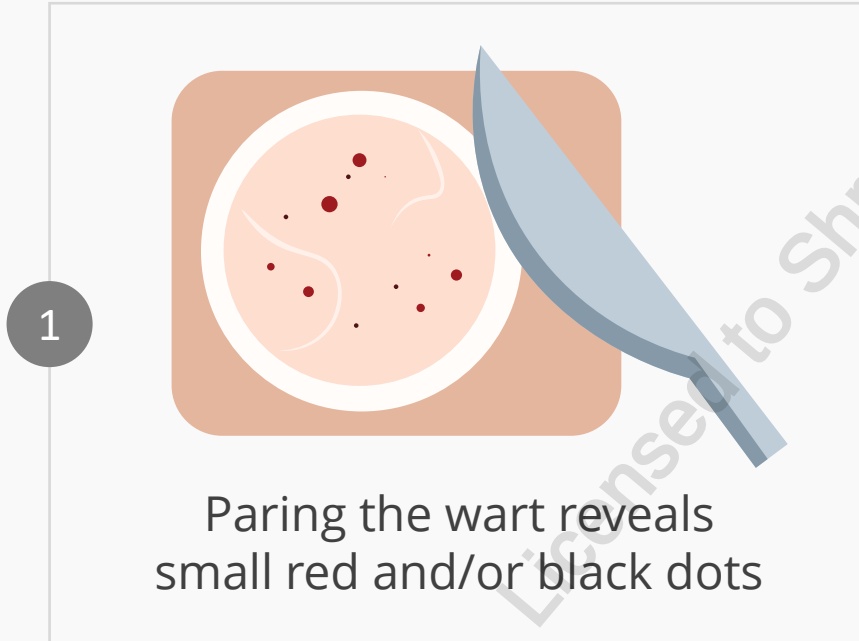
CHARACTERISTICS OF VERRUCAE

TYPE	CHARACTERISTICS
Common warts	• Sharp demarcation • Rough and firm, dome-shaped • Black dots may be present when pared
Plantar warts	• Flat, rough, and thick • Endophytic • Black dots may be present, pinpoint bleeding when surface is pared
Flat warts	• Several, smooth, small papules • Minimally elevated or flat • Pink, brown, or skin-coloured



CLINICAL PRESENTATION

- Assessment of verrucae is typically based on clinical appearance
 - Lab tests and biopsies are generally not required
- Assessment features:



Quiz

Click the **Quiz** button to edit this object

CHECK YOUR UNDERSTANDING

Keeping in mind the pathophysiology and mechanism of transmission involved, type in the space provided what risk factors you think may put an individual at a higher risk of developing verrucae.





RISK FACTORS

Risk factors for the development of verrucae include:

- Age (more common in children and young adults)
- Occupation (e.g. handling raw meat)
- Skin conditions (e.g. atopic dermatitis)
- Immunocompromised status (e.g. HIV infection, organ transplant)
- Contact with an individual with warts (direct/indirect)



DIFFERENTIAL DIAGNOSES



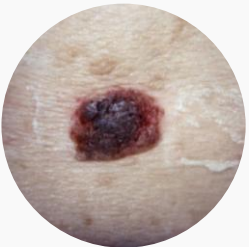
Corns:

Does not normally interfere with skin lines and lacks thrombosed capillaries



Black heel (talon noir):

Does not interfere with skin lines and the black marks are removable



Seborrheic keratosis:

Often pigmented and appear “stuck-on”

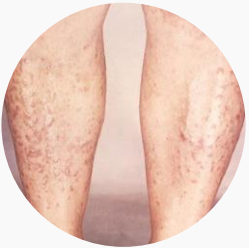


DIFFERENTIAL DIAGNOSES



Acrochordon:

Pedunculated lesion without finger-like projections



Lichen planus and lichen nitidus:

Symmetric distribution and numerous 1 to 2 mm round, smooth papules



Malignancy:

Irregular growth, ulceration, resistant to treatment



WHEN TO REFER

SITUATIONS WARRANTING PATIENT REFERRAL



Examples:

- Warts on face, genitals, or fingernails
- Widespread lesions
- Immunosuppression
- Patient has neuropathy or conditions that are associated with neuropathy (e.g. diabetes, peripheral vascular disease)
- Signs of infection (e.g. pus, pain, redness)
- Failure to respond to treatment
- No symptom resolution after 12 weeks of home treatment



ASSESSMENT

- **Assess symptoms and rule out differential**
 - Are the symptoms and characteristics consistent with verrucae? (e.g. common, plantar, or flat warts)
 - Shape, color, texture, size of lesions
 - Does the lesion interfere with normal skin pattern?
 - Upon paring, are black and/or red dots present?
 - Are the lesions painless or painful when pressure is applied?
 - History and onset
 - Location and distribution of lesions (e.g. localized or widespread, number of lesions, involvement of red flag areas)
- **Patient medical and social history**
 - Medical conditions (e.g. is the patient diabetic, have peripheral vascular disease, or immunocompromised?)
 - Has the patient had any contact with others who have warts?
 - Occupation (e.g. does the patient handle raw meat regularly?)

Quiz

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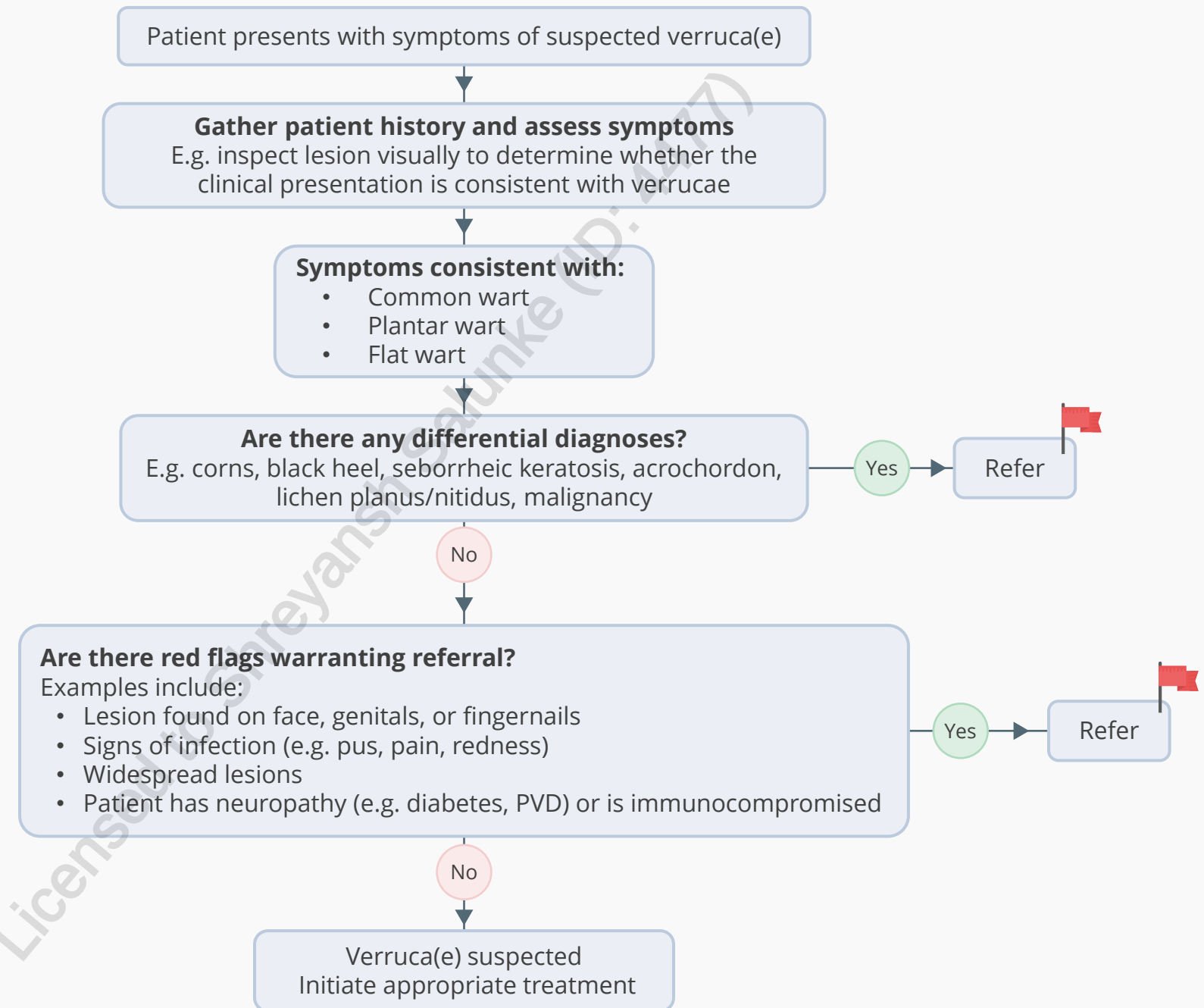
CHECK YOUR UNDERSTANDING

In which of the following cases would a referral be considered or indicated? Select all that apply.

- ☐ **a)** A patient presents with a suspected verruca on the lateral side of the cheek
- ☐ **b)** A patient with a 15-year history of type 2 diabetes presents with a suspected plantar wart
- ☐ **c)** A patient with a history of migraines presents with a lesion that has small black dots
- ☐ **d)** A patient with a 5-year history of hypothyroidism presents with a dark pigmented lesion that has the appearance of being “stuck on” the skin.

ASSESSMENT ALGORITHM

VERRUCAE (VULGARIS AND PLANTAR WARTS)



LEGEND

PVD: Peripheral vascular disease

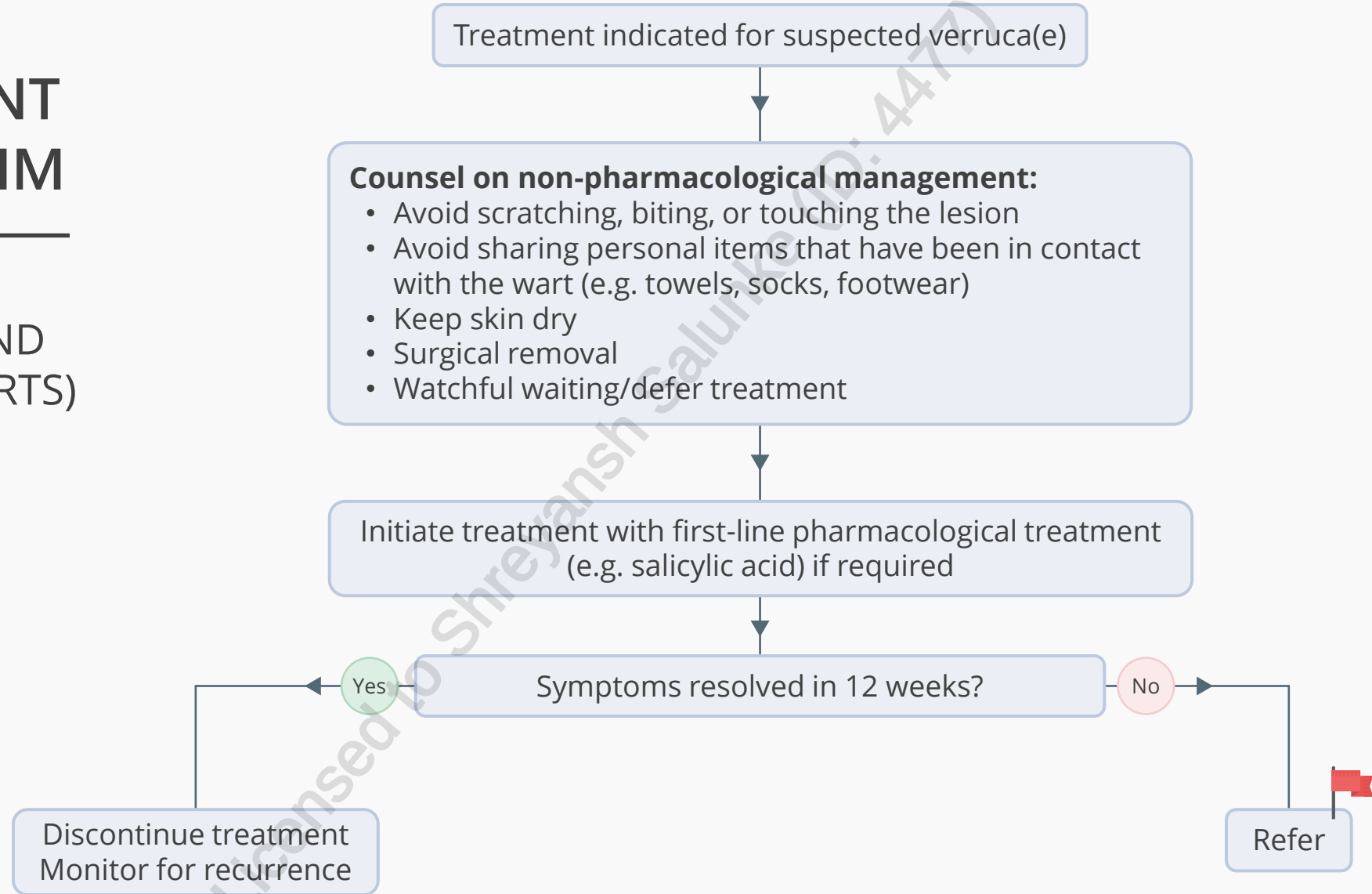
TREATMENT



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TREATMENT ALGORITHM

VERRUCAE (VULGARIS AND PLANTAR WARTS)





NON-PHARMACOLOGICAL TREATMENT

- **Non-pharmacological prophylaxis/management strategies**
 - Avoid scratching, biting, or touching the lesion
 - Avoid sharing personal items that have been in contact with the wart (e.g. towels, socks, footwear)
 - Keep the skin dry and wash hands after touching the wart
 - Remove hyperkeratotic tissue on plantar or common warts (e.g. using pumice stone)
 - Surgical removal
 - Watchful waiting (i.e. defer treatment) for spontaneous resolution
- **Treatment options (pharmacological and procedural)**
 - First-line therapy: salicylic acid, liquid nitrogen cryotherapy



PHARMACOLOGICAL TREATMENT OPTIONS

FIRST-LINE

MEDICATION	ADMINISTRATION AND DURATION	ADVERSE EFFECTS AND CONSIDERATIONS
Salicylic acid (Keratolytic agent)	• Apply product topically to wart • Dosing frequency varies between once daily, twice daily, or every 48 hours depending on concentration and formulation • Duration of treatment up to 12 weeks	• Avoid healthy skin as irritation is common (e.g. may cut medicated plasters to fit wart) • Avoid in patients with neuropathy or conditions that are associated with neuropathy (e.g. diabetes, peripheral vascular disease) • Avoid for facial warts • Prior to application: Soak verruca(e) in warm water for 5 minutes and dry area thoroughly before application
Liquid nitrogen (Cryotherapy)	• Physician administered liquid nitrogen treatment • 1 treatment every 2 to 3 weeks until resolution	• Painful and not typically used for younger children • Pare plantar and hyperkeratotic warts prior to administration • Can be used in combination with salicylic acid • Blistering, pigment changes, tenderness are common • Over-the-counter cryotherapy products may be less effective



PHARMACOLOGICAL TREATMENT OPTIONS

ALTERNATIVE TREATMENTS REQUIRING REFERRAL TO ANOTHER HEALTH CARE PROVIDER

MEDICATION	ADMINISTRATION AND DURATION	ADVERSE EFFECTS AND CONSIDERATIONS
Intralesional bleomycin (Antineoplastic)	- Physician administered 1 mg/mL injected into the wart as a one-time injection with local anesthetic	- Works through its direct cytotoxic effects - Pain upon injection and for up to 48 hours after - Necrosis at site of injection is expected - Avoid in children, pregnancy, immunosuppressed patients, and those with vascular disease
DPCP, DNCB, SADBE (Contact immunotherapy)	- Dermatologist administered - Commonly performed twice weekly for up to 10 weeks	- Pruritus, edema, erythema, and burning are common - Works by inducing a local immune response at the site of the wart
5-fluorouracil (Antineoplastic)	- Applied topically once daily then occluded for 4 to 12 weeks (5% cream) - Can also be injected intralesionally	- Works by preventing proliferation of the wart - Pigment changes, scarring, pain, and burning are common - May cause photosensitivity; patients should use sunscreen
Pulsed laser treatment	Physician administration every 3 to 4 weeks until resolution	- Works through damaging hemoglobin selectively to destroy the vasculature in the wart - Pare wart prior to administration - Can be used in combination with salicylic acid - Pain, blistering, pigment changes, and scarring are common



PHARMACOLOGICAL TREATMENT OPTIONS

ALTERNATIVE TREATMENTS REQUIRING REFERRAL TO ANOTHER HEALTH CARE PROVIDER

MEDICATION	ADMINISTRATION AND DURATION	ADVERSE EFFECTS AND CONSIDERATIONS
Cantharidin (Keratolytic agent)	0.7% solution applied topically then occluded with tape every 3 weeks for up to 4 treatments - Physician-administered	- Blistering within 24 hours is common - Painless during in-office administration and simple to treat multiple warts - Blistering post-administration may cause discomfort and pain
Imiquimod (Immunomodulator)	Apply 5% cream topically to pared warts for up to 12 weeks	- Indicated for genital and perianal warts - Off-label use for verrucae – efficacy not established - Non-scarring and painless application - Skin reactions, mild pain



MONITORING

- Goal of therapy: removal of visible verruca(e), or warts
- Check skin regularly for new warts (e.g. recurrence)
- Patients should notify health care professional if treatment was unsuccessful or if wart grows or spreads during treatment
- Monitor for signs of infection (e.g. pain, swelling, redness, pus)
- Monitor for adverse effects of medications and discontinue if severe irritation, erythema, and/or systemic symptoms arise

Quiz

Click the **Quiz** button to edit this object

CHECK YOUR UNDERSTANDING

Is the following statement true or false?

Topical salicylic acid is an effective treatment that has anti-viral effects to eradicate the HPV virus that causes verrucae.

☐ True

☐ False



CLINICAL PEARLS

- Proper administration is critical to success of the therapy
 - Ensure that detailed instructions and adequate counselling are provided
- A referral may be required if warts are persistent or widespread
- Higher concentrations of salicylic acid are typically used on thicker skin (e.g. soles)
- Current goal of therapy is removal of visible verrucae but does not confirm that HPV is eradicated

PATIENT CASE





CASE BACKGROUND

Background: AS is a 31-year-old woman who presents with a lesion on the sole of her foot. The lesion interferes with skin lines, is the same colour as the skin, and is painless unless pressure is applied to it. This is the first time this has happened to her, and she noticed it yesterday. She has not tried any treatment for this yet and when she picked at it, she noticed black dots.



PAST MEDICAL HISTORY

- Migraines

ALLERGIES

- No known allergies

SOCIAL HISTORY

- No smoking
- No alcohol
- Swims twice weekly at a public pool

MEDICATIONS

- Sumatriptan, as needed
- Propranolol 40 mg, twice daily



CASE QUESTIONS

1. Which ONE of the following is the LEAST appropriate recommendation for AS?
 - a) Do not share personal items such as towels and slippers
 - b) Keep feet dry
 - c) Remove hyperkeratotic tissue by scratching the wart using clean hands
 - d) Do not share tools used to pare the wart

2. Which of AS's following features are consistent with an assessment of plantar warts? Select all that apply.
 - a) Appearance of black dots after picking at it
 - b) Interfering with skin lines
 - c) Painless unless pressure is applied
 - d) Found on the sole of the foot



CASE QUESTIONS

3. Which ONE of the following is the MOST appropriate treatment to start for AS?
- a) Salicylic acid
 - b) Imiquimod
 - c) Pulsed laser treatment
 - d) 5-Fluorouracil
4. Which ONE of the following is INCORRECT regarding salicylic acid counselling?
- a) Apply liberally to the wart and surrounding skin
 - b) Pare the wart prior to administration
 - c) Soak the foot in warm water for 5 minutes prior to administration and dry thoroughly
 - d) May take up to 12 weeks for resolution



CASE SOLUTIONS

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CASE SOLUTIONS

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REFERENCES



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reference list